



Revolutionizing marine Protection: Innovative designs for In-Situ applicable underwater antifouling superoleophobic coatings

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ABSTRACT

There is a high demand for underwater antifouling coatings for marine engineering and underwater activities. However, *In-situ* painting coating in a water environment for various large substrate surfaces is ideal but extremely challenging. In this work, we present a novel approach to achieving a fouling-repellent superoleophobic coating that can be directly applied underwater. The coating was prepared facilely via one-pot synthesis using epoxy resin E51 and 2-hydroxyethyl methacrylate as main components. This method involves the hydrolytic migration of side chains (Trimethylsilyl Methacrylate) interlocked with covalent cross-linking, enabling direct adhesion to a variety of wet substrates underwater without the need for surface pre-treatment. The resulting coating exhibits both underwater superoleophobicity (underwater oil contact angle > 150°) and versatile, sustainable substrate adhesion with a shear strength of up to 50 kPa, making it suitable for various underwater applications, such as oil/water separation. This work represents a facile and novel paradigm for designing future underwater antifouling coatings with superoleophobic properties and direct underwater adhesion.

1. Introduction

In the marine industry, various underwater equipment[1] and structures[2] (e.g., ships, drilling platforms, submarine pipelines, bridge piles, etc.) are generally submerged in water for extended periods. Unfortunately, they frequently suffer a significant reduction in their operational lifespan due to oil fouling, biofouling and corrosion caused by adhering organisms.[3] These phenomena not only diminish their longevity but also escalate operational costs. Therefore, utilizing antifouling coatings is the most convenient and economical method to prevent the above-mentioned fouling and corrosion.[4–7].

Various antifouling coatings have been designed and prepared, such as self-polishing antifouling coatings[8], fouling release coatings[9], and dynamic surface antifouling coating.[10] However, these materials often lack superoleophobic function due to their hydrophobic nature, resulting in only modest antifouling performance.[11–15] In contrast, many marine organisms hold superhydrophilic biological surfaces, such as fish and algae, exhibiting excellent functions of antifouling, oil

repellency, anti-bioadhesive, and drag reduction properties underwater. Yet, synthetic materials face challenges in achieving high strength and long-term adhesion underwater due to their highly hydrophilic structure and hydration layer interface, which also leads to that many antifouling coatings are required to be coated in the air and cannot be directly underwater *in situ* proceed, due to the highly hydrophilic material components easily dissolving in water.

Existing strategies for fabricating underwater superoleophobic antifouling surfaces involve complex pretreatments of substrates and detailed manipulation of chemistry and macro/nano structures, typically performed in air or dry conditions.[16–19] Our previous study designed an amphiphilic polyurethane coating with superoleophobicity and adhesion to various substrates underwater.[20,21] However, these materials lack applicability for direct underwater painting and wetting adhesion, due to the hydrophilicity of the precursor and the resulting coating. It is apparent that while the hydrophilicity of coating materials contributes to effective underwater superoleophobicity, it poses a hindrance to direct underwater painting. Thus, addressing maintenance

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issues for marine equipment requires developing a simple and universal strategy to enable coatings to be directly painted underwater while maintaining superoleophobicity and substrate adhesion. However, achieving this goal presents a significant challenge.

In this study, we present the complementary roles employed in designing *in-situ* applicable underwater antifouling superoleophobic coatings. 1) Using a hydrophobic polymer system is critical, as hydrophobicity is essential to prevent the dissolution and consumption of the polymer components in a water environment. In addition, achieving appropriate viscosity of the polymer solution facilitates substrate wetting and film forming during the coating process. 2) Simultaneous affinity to a substrate and little water has been designed to eliminate the hydration layer of the substrate, thus achieving robust and versatile underwater bonding without the need for any pretreatments. 3) Importantly, water-responsive units have been incorporated into the system, allowing the coating surface to spontaneously self-reverse to superhydrophilic, ultimately achieving the desired underwater

superoleophobic antifouling surface function. By considering these designing factors, we have developed coating materials that can be readily applied to various underwater substrates without any pretreatment, thereby providing outstanding underwater antifouling superoleophobic performance combined with strong substrate bonding and corrosion protection.

2. Results and discussion

2.1. Design and preparation of superoleophobic coatings for underwater application

To achieve *in-situ* applicable underwater antifouling superoleophobic coatings, we designed a strategy based on the aforementioned three rules. Specifically, hydrophobic trimethylsilyl methacrylate (TBSM), excess epoxy resin E51, m-xylylenediamine (MXDA), isophorone diisocyanate (IPDI), and hydrophilic 2-hydroxyethyl

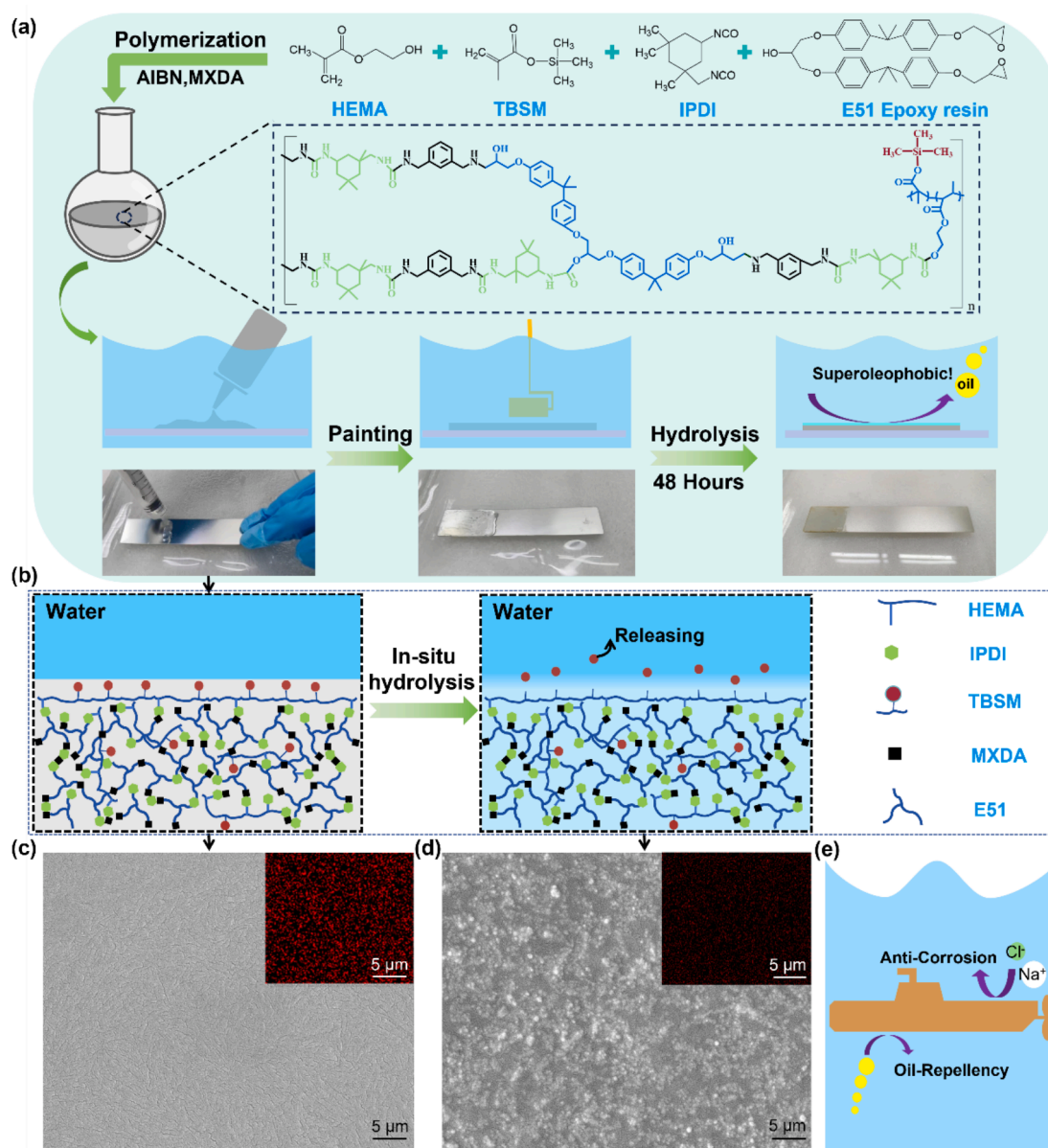


Fig. 1. a) Schematic illustration of the preparation process of the underwater superoleophobic coating formed by *in-situ* hydrolysis. b) Interfacial schematic of the coating forming superoleophobic surface via *in-situ* hydrolysis underwater. SEM photographs of the coating surfaces c) before and d) after hydrolysis (the inset showed the silicon content determined by EDS). e) Schematic illustration of the potential underwater applications of the coating for anti-oil fouling and anti-corrosion.

methacrylate (HEMA) were first dissolved in water-soluble dimethylacetamide (DMAc) and then polymerized via a free radical polymerization reaction with 2,2'-Azobis(2-methylpropionitrile) (AIBN) as the initiator to obtain a wetting adhesive amphiphilic polymer (WAP) solution (Fig. 1 and Fig. S1). Gel-permeation chromatography (GPC) results confirmed the successful preparation of WAP, with the molecular weight of the WAP being about 10 kDa (Fig. S2), leading to a suitable viscosity. Thus, when the WAP solution was extruded directly underwater onto the substrate without any substrate surface pretreatment and then scraped flat to obtain a coating with a smooth surface, the amphiphilic segments self-migrated to the substrate to eliminate its hydration layer. As shown in Fig. S3, the viscosity of the WAP solution increased significantly with reaction time, forming highly viscous polymers for underwater adhesion. Fourier transform infrared (FTIR) spectroscopy indicated that the $-NCO$ groups were fully consumed, as evidenced by the disappearance of their characteristic peak (Fig. S4), indicating the coating continuously reacted with the substrates. The storage and loss modulus of the coatings confirmed the successful preparation (Fig. S5). It is noteworthy that TBSM segments on the coating surface could contact water molecules and undergo underwater in-situ hydrolysis to generate hydrophilic carboxyl groups through the hydrolysis of silicon-oxygen bonds. As a result, the surface of the WAP coating spontaneously self-reversed to superhydrophilic, achieving the

desired underwater superoleophobic antifouling surface function. As shown in Figs. S4 and S6, TBSM in water were hydrolyzed to form a white layer, and the FTIR spectra of the coating after hydrolysis exhibited a new peak at 1600 cm^{-1} , assigned to the carboxyl group. [22,23] The X-ray photoelectron spectroscopy (XPS) (Fig. S7) and Raman spectroscopy (Fig. S8) of the coating surface also confirmed the hydrolysis that the silicon content and methyl groups decreased after hydrolysis. The results of scanning electron microscopy (SEM) and energy dispersive spectrometry (EDS) demonstrated that the surface became rough and the silicon element content decreased significantly after immersion for 7 days, due to the hydrolysis of the silyl ester groups on the surface, as shown in Fig. 1. Thus, with the aid of the in-situ generated hydrophilic groups, the coating interacted with water molecules to form a hydrated surface, leading to the in-situ formation of an underwater superoleophobic surface (Fig. 1a).

2.2. Directly underwater wetting adhesion on various substrates

It is highly desirable that the polymer solution can be applied directly underwater onto various substrates without requiring surface treatment to form an advanced functional coating, due to the ease of on-site application without complex procedures and equipment requirements. Thus, the ability of the coatings to be directly applied

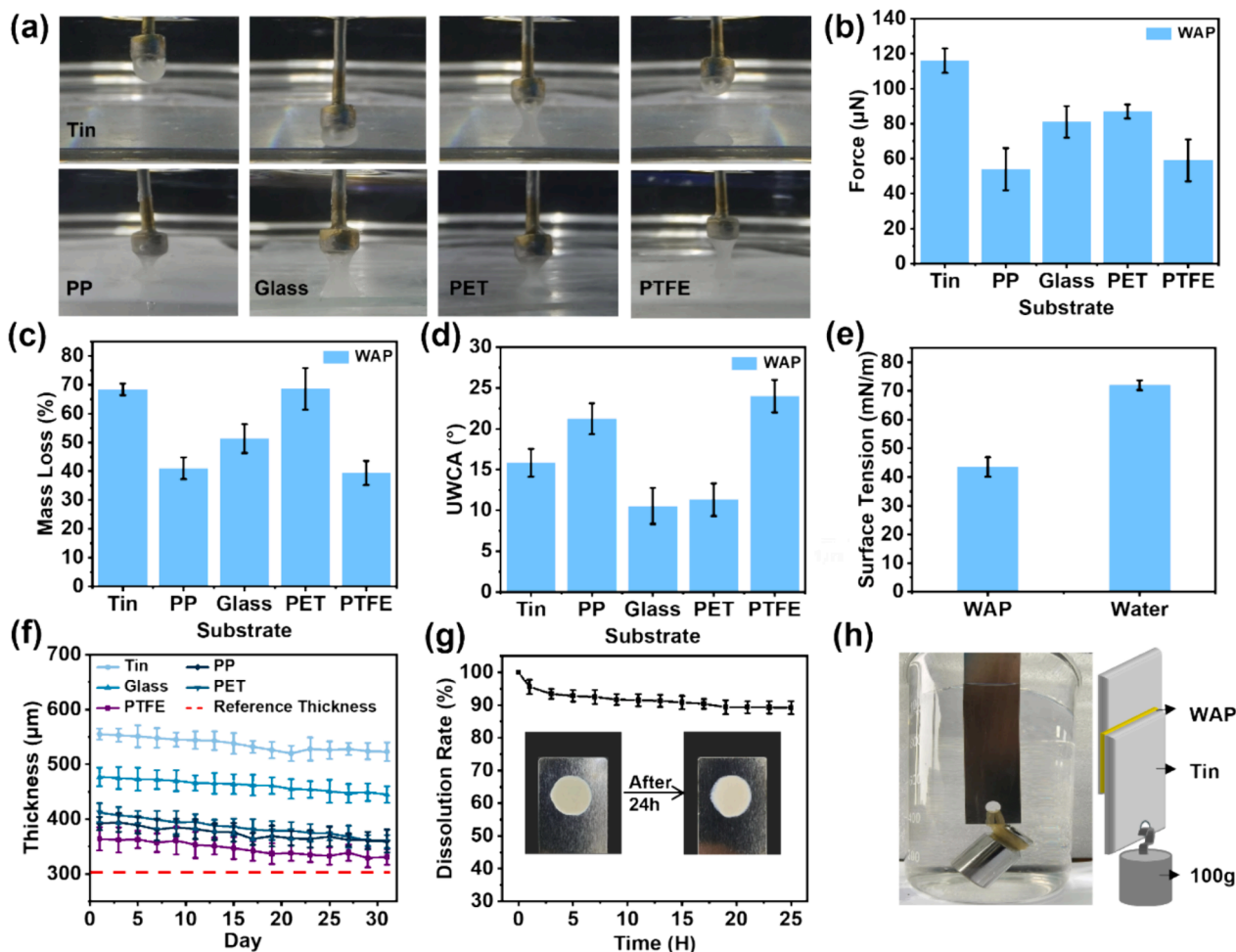


Fig. 2. Underwater coating capability. a) Schematic illustration of the underwater adhesion process of the WAP solution to various substrates. b) Schematic graph of adhesion force of WAP on various underwater substrates. c) Schematic graph of mass loss at the end of the adhesion process of WAP on various underwater substrates. d) Schematic graph of contact angles of WAP on various substrates in a water environment. e) Schematic graph of surface tensions of the WAP solution and water. f) Thickness variation of the coating on various substrates underwater, minimum thickness requirements refer to < NACE SP0108-2008 > . g) Schematic graph of the dissolution rate of the WAP solution for 24 h. h) Schematic illustration of the coating on a Tin hanging weight of 100 g underwater for 12 h. (adhesive area 25 mm × 25 mm).

underwater on a variety of substrates without requiring surface treatment was evaluated, as shown in Fig. 2a and Movie S1. This WAP solution could successfully adhere to the untreated surfaces of various substrates, such as tinplate (Tin), polypropylene (PP), glass, polyethylene terephthalate (PET), and polytetrafluoroethylene (PTFE), directly underwater with strong adhesion, noticeable stretching behavior was observed. The corresponding adhesion force was further quantitatively determined, as well as the mass loss of the WAP solution (Fig. 2b-c), and industrial E51 epoxy resin was used as a control (Fig. S9). As shown in Fig. 2b, the adhesion force of the WAP solution instantly adhered to Tin was about 120 μN , and 80 μN on hydrophilic glass, as well as 60 μN on the low-surface-energy polytetrafluoroethylene (PTFE). As expected, when the WAP droplet was lifted to leave the substrate surface, some WAP remained on the substrates due to the high adhesion ability. Thus, the mass loss of WAP was determined. As shown in Fig. 2c, obvious mass loss was observed: the WAP lost a mass of 70 % on Tin and 40 % on PTFE. Such good underwater adhesion was further confirmed by the underwater wetting ability using underwater contact angles (UWCA). When 5 μL of WAP was added to different substrates, the UWCA of all samples was below 30° (Fig. 2d). Furthermore, the UWCA of WAP was lower than that of pristine E51, indicating the excellent underwater substrate wetting and spreading performance of the WAP (Fig. S9c). To further demonstrate the underwater wetting and spreading properties of the WAP solution on the substrates, the surface tension of the solution needs to be measured. The surface tension of a solution that can be directly applied underwater should be lower than that of water so that it can wet the surface of the substrate well in water. As shown in Fig. 2e, the surface tension of the WAP solution was about 40 mN/m, lower than that of water (70 mN/m). These results confirmed the good instant underwater adhesion of the WAP solution to

various substrates without surface treatment. Due to the excellent adhesion, the WAP could adhere to various substrates and remain immersed underwater for 30 days without falling off, and the thickness was slightly decreased but always higher than the recommended thickness of 300 μm (relevant coating thickness standards, NACE SP0108-2008), indicating excellent underwater adhesion stability (Fig. 2f). The thickness change may be due to the release of DMAc solvent inducing the coating shrinking. To further confirm the low dissolution of the WAP solution in water, the weight of the Tin with 0.5 g of WAP solution was recorded, and the results showed that only about 10 % of its weight was lost after 24 h, indicating the loss of DMAc and the absorption of water, slightly higher than its solid content of 80 %.

To further evaluate the excellent adhesion of the WAP coating, the adhesion strength of the coating on various substrates, including Tin, steel, aluminum (Al), PET, and PP, was determined by shear testing (Fig. S10). The results demonstrated the good adhesion strength of the coating after underwater painting on various substrates for 48 h, with the shear strength of the coating being about 50 kPa on Tin, about 43 kPa on Al, and about 20 kPa on PP. Importantly, the shear strength on different substrates could be maintained for many days and gradually decreased to 21 % of the original strength after 1 month of underwater immersion. Thus, based on the excellent adhesive strength of the coating to the substrate, a 100 g weight could be hung underwater stably for 12 h (Fig. 2h).

2.3. Underwater superoleophobicity with excellent stability

With the aid of the *in-situ* formation of the hydrophilic surface, the coating exhibits a hydrophilic effect in the air (Fig. S11), not only confirming the presence of the hydration layer but also demonstrating the

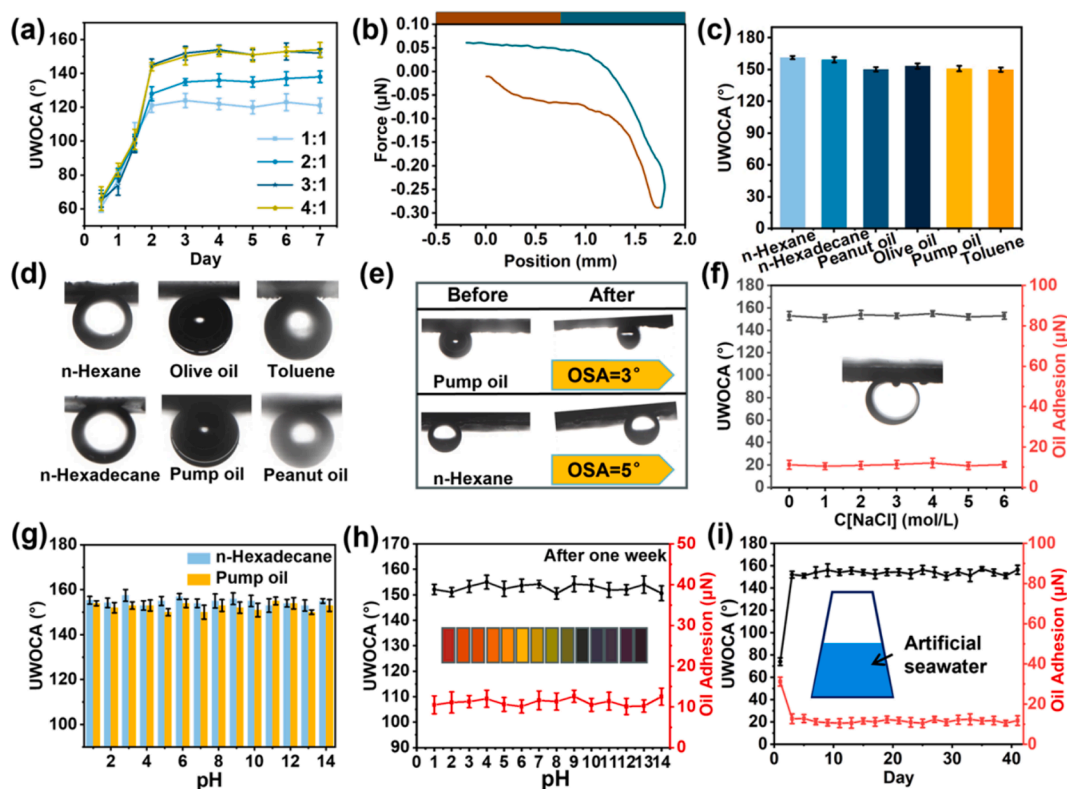


Fig. 3. Underwater superoleophobicity with supreme stability. a) Impact of different TBSM and HEMA ratios on the oil-repellent effect of the coating. b) Evolution of adhesion forces with the preloading of an n-hexadecane droplet on the coated surface. c) Superoleophobicity with different types of oils. d) Demonstration of superoleophobicity of different oils on the underwater coating surface. e) Various moments when a pump oil droplet or n-hexadecane droplet quickly slid off the coating surface. f) Surface oil adhesion and underwater oil contact angle (UWOCA) after immersing the coating in 0–6 mol/L NaCl solution for 1 day. g) UWOCA and surface oil adhesion after immersing the coating surface in solutions with different pH values (1–14) for 1 day. h) UWOCA after immersing the coating surface in solutions with different pH values (2–14) for one week. i) UWOCA and surface oil adhesion of the coating surface after immersion in artificial seawater for 40 days.

underwater oil-repellent properties of the coating. As shown in Fig. 3a, the superoleophobicity was increased gradually with the immersion time increasing and the UWCA was increased from 60° to $> 150^\circ$. Moreover, it is observed that the high quantity of silicon-based ester groups was helpful for the oil-repellent performance of the coating, and it was determined that the optimal and most economical polymerization molar ratio of TBSM and HEMA was 3:1.

To confirm the underwater oil-repellent ability, the underwater adhesion force of oil droplets on the WAP-coated substrates was measured using n-hexadecane as a model liquid. During the testing process, the oil droplets were compressed and deformed, then quickly returned to their original shape. As shown in Fig. 3b, the oil showed just a low adhesion force of $\sim 10 \mu\text{N}$ on the WAP-coated Tin, confirming the coating's excellent underwater anti-oil adhesion performance. To further illustrate the superoleophobicity of the coating, the underwater oil contact angle (UWOCA) of various oils, such as n-hexane and pump oil, was measured on the WAP-coated Tin. All oil droplets, including low-viscosity oil (such as n-hexane) and high-viscosity oil (such as pump oil), exhibited a good spherical shape on the coating surface with a UWOCA exceeding 150° (Fig. 3c-d). [24] Moreover, various oils exhibited excellent underwater sliding behavior on the coating (Fig. S12), such as the underwater oil sliding angle (OSA) of low-viscosity n-hexane being only 3° , while the OSA of high-viscosity pump oil sliding underwater was just 5° (Fig. 3e). Thus, the WAP coating exhibited an excellent underwater OSA ($< 10^\circ$) (see Movie S2).

Besides good anti-fouling ability, good underwater stability is also vital for underwater superoleophobic coatings, as their hydration layer is often easily damaged during underwater application. Thus, achieving a stable superoleophobic coating is a crucial yet challenging task. [25,26] In this work, the excellent stable superoleophobicity of the WAP coating was evaluated by immersion in NaCl solution with various concentrations (0–6 mol/L) for 24 h, and the results showed that the coating could maintain a high UWOCA exceeding 150° and low oil adhesion force (Fig. 3f). To further confirm the underwater stability of the coating, it was immersed in solutions with various pH conditions (pH: 1–14) for 24 h. The results indicated that the UWOCA of low-viscosity oil (n-hexadecane) and high-viscosity oil (pump oil) on the coating surface still exceeded 150° (Fig. 3g). Moreover, even after immersion in solutions with different pH conditions for one week, the coating maintained a high UWOCA and low oil adhesion (Fig. 3h). These results confirmed that the superoleophobic property of the WAP coating exhibited excellent underwater stability against various NaCl concentrations and extreme acidic and basic conditions. To further confirm the applicability of the WAP solution and coating, the WAP solution was directly painted on the Tin under artificial seawater and then left in the artificial seawater for 40 days. As shown in Fig. 3i, the WAP solution adhered to the tinplate with a relatively low UWOCA and high adhesion force. Thereafter, with hydrolysis and the in-situ formation of a hydration layer, the UWOCA improved to exceed 150° , and the oil adhesion force decreased, demonstrating excellent stability during 40 days of immersion in artificial seawater (Fig. 3i).

Additionally, good mechanical stability is also important for coatings used underwater, such as in lakes, rivers, oceans, and for oil/water separation. To simulate the scouring of sand particles on the coating in a river, a coated tinplate was fixed in 1 L of water containing 10 g of fine sand, and the water was stirred at a high speed of 500 rpm for 24 h. As shown in Fig. S13, no significant changes were observed on the coating surface or its superoleophobic performance. To further demonstrate its mechanical stability, the coating also underwent water droplet and sand particle impact tests, where 5000 drops of water and 30 g of sand were dropped from a height of 50 cm and impacted the coating (Fig. S14). The results also demonstrated no significant changes in the coating surface or its underwater superoleophobicity, consistent with the result of SEM images of the coating after abrasion (Fig. S15). These results confirmed its excellent mechanical stability. [27–29].

2.4. Anti-crude oil fouling, oil/water separation and anti-corrosion

Based on the excellent underwater superoleophobic and substrate adhesion of the coating, it could impart effective crude oil anti-adhesion properties to the substrate. [30] In the air, when a tinplate was poured with crude oil, the uncoated part of the Tin exhibited serious crude oil fouling and could not be flushed clean using water due to the excellent strong adhesion ability of the crude oil, while the coated part exhibited weak crude oil adhesion, leading to crude oil easily flowing down, and could be easily rinsed clean with water (Fig. 4a and Movie S3). More interestingly, the coated Tin could self-clean the crude oil just by direct immersion into the water where the crude oil was quickly leaving without residue (Fig. 4b), confirming the excellent anti-crude oil adhesion performance. [31] Based on the excellent anti-crude oil performance and the hydrophilic hydration layer of the WAP coating, the WAP-coated double-layer nylon mesh was used as a membrane for oil/water separation. [32] As shown in Fig. 4c-d and Movie S4, the oil/water mixture could be quickly separated via the membrane that the blue-labeled water could pass through the coated mesh with the interaction with the hydrophilic hydration layer, while yellow-labeled oil was repelled by the coated mesh, with high oil/water separation fluxes (Fig. S16) and separation efficiency (Fig. S17).

In marine environments, underwater coatings should not only possess anti-fouling properties but also exhibit certain corrosion resistance to protect the substrate from seawater corrosion. The corrosion resistance performance of the coating is equally important. To demonstrate the corrosion resistance of the coating, a coated Tin and an uncoated one were immersed in artificial seawater for 30 days to observe the corrosion and record the impedance modulus curves, as shown in Fig. 4e-f. [33] The surface of the coated tinplate sample was intact without obvious change, while the uncoated one was severely corroded (Fig. 4e), which was also confirmed by the SEM images (Fig. S18). Interestingly, when the coating was scraped off, the surface of the substrate was clean and not corroded even after 30 days, confirming the good anti-corrosion of the coating. The impedance modulus of the uncoated tin and the coated one were $2.2 \text{ k}\Omega$ and $1.35 \text{ M}\Omega$, respectively, increasing 1000 times, confirming the coating's superior anti-corrosion performance. To further illustrate the corrosion protection effect of the coating, the polarization curves of coated and uncoated tin immersed in artificial seawater for 14 days were recorded. It can be observed that the polarization curves of the coated sample shifted towards more positive corrosion potentials, and the corrosion current was also smaller, compared to the uncoated one (Fig. 4g). These results indicated that the coating also could effectively protect the substrate from the corrosion under seawater. [34,35].

3. Conclusion

In summary, based on the novel mechanism of group hydrolysis and a synergistic segment-oriented/covalent anchoring strategy, superoleophobic coatings can be readily prepared without the need of any surface treatment on various substrates. These coatings demonstrate exceptional long-term superoleophobicity when applied directly underwater onto wetting substrates, even in various extreme environments, such as highly acidic or alkaline solutions and artificially simulated seawater. In addition, these coatings serve multiple functions including oil/water separation, self-cleaning of crude oil contamination, and corrosion protection. This work not only demonstrates an innovative example of underwater painting directly on wetting substrates underwater, but also opens a new avenue for designing facile surface-functional materials. With its straightforward preparation and coating process and outstanding comprehensive performance, this strategy offers fresh perspectives for the future development of long-term, robust, and direct underwater adhesion and antifouling coatings.

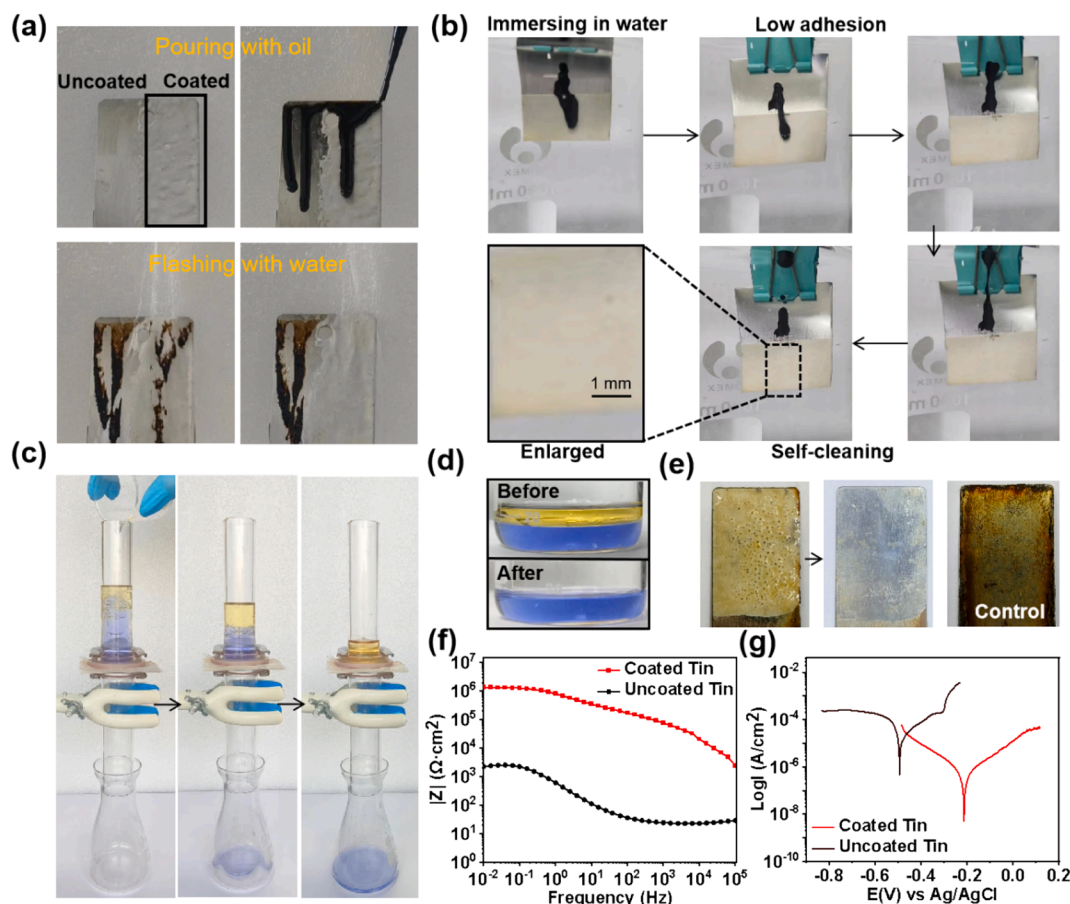


Fig. 4. Underwater Anti-Adhesion Application. a) Effective anti-crude oil adhesion capability of the coating on the right side of the tinplate. b) Coating surface with self-cleaning ability. c) Oil/water separation process of nylon fabric after immersion in the product. d) Presentations before and after oil/water separation. e) Demonstration of the anti-corrosion protection effect of the coating on a tinplate. f) Bode plots of uncoated and coated tin after being immersed in artificial seawater for 30 days. g) The potentiodynamic polarization curves of uncoated and coated tin after being immersed in artificial seawater for 14 days.

4. Experimental section

4.1. Materials

Dimethylacetamide (DMAc) was purchased from Tianjin Zhiyuan Kemei Chemical Co., Ltd. and dehydrated using 4 Å molecular sieves before use. Trimethylsilyl methacrylate (TBSM), 2, 2'-azobis (2-methyl propionitrile) (AIBN), isophorone diisocyanate (IPDI), epoxy resin E51, dibutyltin dilaurate (DBTDL), *m*-diacetyl diamine (MXDA), hexane, *n*-hexadecane, 2-hydroxyethyl methacrylate (HEMA) were acquired from Shanghai Macklin Biochemical Co., Ltd. and used as received. Pump oil was purchased from Huaxin Chemical Reagent Co., Ltd. Olive oil and peanut oil were purchased from local supermarkets and used as received. Crude oil (Viscosity: ~300 cP) was provided by PetroChina.

4.2. Instruments and characterization

FTIR spectra were obtained at room temperature to confirm the successful synthesis of the product using a TENSOR II+Hyperion 2000 (Bruker, Germany). The viscosity of the product was measured by a rheometer (MCR92, Anton Paar, Germany). The average molecular weights (M_n and M_w) and polydispersity were measured by gel permeation chromatography (GPC) (1515, Waters, USA). Underwater oil contact angle and air-water contact angle were measured at ambient temperature using a contact angle goniometer (JC2000A, Shanghai Zhongchen Force Measuring Instrument Co., Ltd., China). The microscopic morphology of the coating surface was evaluated using a scanning electron microscopy (Gemini 300, Zeiss). The X-ray photoelectron

spectroscopy (XPS) of the coating surface was evaluated using a X-ray photoelectron spectrometer (ESCALAB 250Xi, Thermo Scientific). The Raman spectra of the coating surface was evaluated using a Raman spectroscopy (DXR3xi, Thermo Scientific). The storage (G') and loss (G'') moduli of the hydrogel coatings measured by a rheometer (MCR92, Anton Paar, Germany). The underwater adhesion of the coatings to *n*-hexadecane (20.0 μL) and the adhesion of the products to various substrate surfaces underwater were evaluated with a tensiometer (K-100, Bruker, Germany). To assess the corrosion resistance of the coating, use an electrochemical workstation (GAMRY Reference 3000, USA).

4.3. Preparation of coatings

TBSM (1.2 g, 7.594 mmol), HEMA (0.34 g, 2.538 mmol), E51 (2.4 g, 3.125 mmol), and IPDI (1.2 g, 5.405 mmol) were dissolved in 1.2 g of DMAc, reacted as precursors for 2 h at 80 °C, and stored at room temperature. Then, 5 g of precursor and 0.25 g of MXDA were mixed and stirred at room temperature for 1 day to obtain a homogeneous polymer solution. The polymer solution was applied underwater to different substrates and then cured in water for 2 days to form the final coating.

4.4. Underwater coating capability

To evaluate the underwater coating performance of the product, a syringe containing 1.0 mL of the product was first extruded underwater on a tin plate and then flattened with a squeegee to obtain a coating. The viscosity of the product was measured using a rheometer (MCR92, Anton Paar, Germany) at different cross-linking times. The adhesion and

surface tension and mass loss of the products to various substrate surfaces were evaluated underwater using a tensiometer (K-100, Bruker, Germany). The thickness of the coating on different substrates was measured using vernier calipers; the contact angle of the product and E51 epoxy resin on different substrates underwater was measured using a contact angle goniometer (JC2000A, Shanghai Zhongchen Force Measuring Instrument Co., Ltd., China). In order to evaluate the dissolution rate of WAP, the mass of the tinplate was first weighed using an electronic balance, then a syringe containing 0.5 g of WAP was extruded on the tinplate and placed in water, and the plate was removed at regular intervals and the water was wiped off from the surface before being weighed again.

4.5. Underwater superoleophobicity

The product coated tin plate was immersed in a deionized water bath. To measure the underwater contact angle of the various oils on the product-coated tin plate, a syringe needle containing 5.0 μL droplets of various oils (n-hexadecane, n-hexane, toluene, peanut oil, pump oil, and olive oil) was placed under the coating and the droplets were injected, and the oils were then uplifted by buoyancy to contact the surface of the coating. The corresponding underwater adhesion of the coating to n-hexadecane (20.0 μL) was evaluated using a tensiometer (K-100, Bruker, Germany).

4.6. Substrate adhesion tests

For lap shear adhesion, test samples were prepared by first applying the solution underwater to a substrate slide and then covering it with another substrate slide. The coated slide was immersed in water for 24 h to obtain the sample. The 180° peeling test sample was made in the same way as the lap-shear adhesion sample. The samples were evaluated using a universal testing machine (HZ-1007E, Lixian Instrument Technology Co., Ltd., Dongguan, China) at a speed of 100 mm/min. The tensile force was applied to the substrates until separation occurred between the substrates and then the maximum stress and strain values were recorded. The shear strength is the ratio of the maximum stress to the initial coating area.

4.7. Oil/water separation performance

To evaluate the oil/water separation performance of the product, an oil/water separation device was first constructed, and then two overlapping pieces of 5 cm x 5 cm nylon gauze were immersed into the product, and after being completely submerged, the two pieces of gauze were taken out and immersed into the water, so that the product on the gauze was fully hydrolyzed. After waiting for 3 days, the gauze will be clamped in the oil/water separation equipment and the mixture of peanut oil and water will be poured in to start the test until the end of the water flows down completely. After testing the gauze for oil–water separation using peanut oil, it was then tested for flux through the cycle. Water was poured into the oil–water separation device and the mass of water passing through in 10 s was weighed while keeping the height of the water constant, and then the flux was calculated.

4.8. Mechanical stability

To evaluate the mechanical durability of the coating on a tinplate, sand-flow impact tests in water, water droplet impact tests, and airborne sand-flow impact tests were conducted. The method for the sand-flow impact test in water is outlined as follows: Firstly, the tinplate coated with the coating is immersed in a beaker containing 1 L water and secured in place with a clamp. Then, 10 g sand grains with an average diameter of approximately 300 μm are poured into the water, and the stirrer is activated. To simulate the impact of sand grains in water, the stirring speed is maintained at 500 revolutions per minute (equivalent to

5.24 m/s) for 24 h. The water droplet impact test involves releasing approximately 5000 water droplets (each about 20 μL) continuously from a height of 50 cm onto the surface of the tinplate coated with the coating, with the process lasting approximately 54 min. Additionally, the coating surface is impacted by 50 g of sand grains with an average diameter of approximately 300 μm from a height of 50 cm for 30 s.

4.9. Corrosion resistance testing

Using a tinplate sample (working electrode) along with a reference electrode and an electrolyte, construct an electrochemical testing setup. The samples were immersed in a 3.5 % NaCl solution for 24 h before testing the low frequency impedance properties of the samples. Ensure that the working electrode and reference electrode are properly connected to the electrochemical workstation (GAMRY Reference 3000, USA) and maintain a constant distance and position between the electrodes. Impedance performance tests were first performed to obtain the bode plots of the coated samples and the control after 30 days of immersion. Then, conduct polarization curve tests to obtain the polarization curves, as well as the corresponding polarization potential and polarization current, for the tinplate coated with the coating and the uncoated tinplate after immersion for one week.

CRediT authorship contribution statement

Zhaoji Xia: Methodology, Investigation, Formal analysis, Data curation. **Xiubin Xu:** Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis, Conceptualization. **Xusheng Jiang:** Methodology, Investigation. **Jianmin He:** Methodology, Investigation. **Yanting Chen:** Methodology, Investigation. **Danfeng Yu:** Methodology, Investigation, Funding acquisition. **Zhenyu Wang:** Methodology, Investigation. **Xu Wu:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization. **Hongbo Zeng:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.154157>.

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